An Oblique-Robust Absolute Visual Localization Method for GPS-Denied UAV With Satellite Imagery

Yuan Chen and Jie Jiang

***Abstract*— Localization based on onboard cameras and satellite images is a promising solution for unmanned aerial vehicles (UAVs) when the global position system (GPS) is unavailable. However, apart from the variations in illumination and ground surface, the rotation and viewpoint differences between UAV images and satellite images are substantial due to nonperpendicular camera angles and nonnorthward orientations, making matching difficult. To overcome these challenges, this article presents a novel UAV localization method by matching oblique UAV images and ortho-photograph satellite imagery. First, a lightweight SE(2)-steerable network (SeS-Net) is designed to extract rotation-equivariant dense features from images. Then, an improved peakness measurement is introduced to generate a high-response detection map, from which the discriminative sparse features encoded rotation-invariance for matching are derived. Lastly, camera poses are estimated and refined with the proposed local-to-global matching strategy. This strategy locates the local area related to the map and removes outliers by clustering the matches with a Gaussian model, followed by the adaptive high-quality match selection (AHMS) process to mitigate the impact of local nonlinear distortion on pose estimation. The absolute visual localization (AVL) framework we establish exhibits strong generalization and is applicable to other scenes without additional training. Multiple image matching and UAV localization experiments demonstrate that the proposed method accesses high robustness to drastic changes in perspective and ground surface. Furthermore, its precision and stability in localization surpass both conventional and other deep learning methods.**

***Index Terms*— Absolute visual localization (AVL), image matching, satellite imagery, steerable convolution network, unmanned aerial vehicle (UAV) image.**

I. Introduction

With the development of unmanned aerial vehicle (UAV) technology, the agility of UAV and its convenience of capturing images contribute to more extensive applications such as inspection, mapping, surveillance search, photography, and combat [1]. To fulfill these complex tasks, reliable autonomous localization and navigation of UAVs are required. Global Position System (GPS) and inertial navigation system (INS) are commonly used, while in some challenging situations such as military actions, operations between urban buildings, and planetary exploration, GPS signals may be unavailable or interfered with. The error accumulates over time within inertial measurement units (IMUs) making INS unable to provide reliable long-term navigation. Image sensors are usually equipped as the "eyes" of the UAV to complete various tasks. Localization through image matching with a preexisting database offers the benefits of low cost, compact size, and independence from supplementary hardware equipment and received signals. Therefore, the method of vision-based localization and navigation has become a beneficial scheme to replace or supplement GPS and INS.

The technology of UAV visual localization is mainly divided into relative visual localization (RVL) and absolute visual localization (AVL) [1]. The utilization of consecutive frames for RVL, such as visual odometry (VO) and simultaneous localization and mapping (SLAM), also encounters a similar drift problem as in INS, and neither of them provides georeferenced coordinates without additional information including velocity and scale [2]. AVL locates UAV with the georeferenced reference data, and there is no correlation between pose errors of different positions at different times. These methods match UAV images with remote sensing data, utilizing geospatial information from Geographic Information Systems (GISs), such as terrain [3], [4], [5], distribution of landmarks [6], [7], and road networks [8], [9], [10]. The information in GIS is more difficult to acquire compared to 2-D satellite images from which it is derived, and has lower resolution and slower data updates. Therefore, matching with satellite images directly is more practical and efficient [11], [12].

UAV images are remote sensing images taken at low altitudes with high resolution, large scale, and real-time capability. When matching with satellite images, it is necessary to overcome extreme differences in scale, rotation, and illumination, as well as changes in object appearance over time [2]. Most previous works focus on top-down view image matching [11], [13], [14], which assume UAV images as nadir view or orthoproject them with IMU. However, during real UAV flights, it is difficult to keep a fixed perspective due to the influence of airflow, and a nonnadir view may be needed for other image tasks [15]. In oblique images, occlusion and nonlinear deformation caused by the terrain

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are challenging, and few algorithms are able to tackle the issues above. As conventional methods are susceptible to scene changes, some researchers have employed neural networks to learn to estimate the geometric transformation relationship between UAV and satellite images and obtain the poses of UAVs. For example, Goforth and Lucey [12] adopted a deep Lucas– Kanade network to align two images and estimate the homography. Shetty and Gao [16] built two networks to estimate the position and attitude of UAV, respectively, but the accuracy and generalization are suboptimal. To address the matching challenges between oblique UAV images and satellite images, this article proposes a UAV-satellite image matching method based on the steerable feature extraction network. High-quality matching pairs for pose estimation are adaptively selected during feature matching, followed by UAV geo-localization with the Perspective-n-Point (PnP) algorithm under GPS-denied conditions. The main contributions of this article are as follows.

1) A lightweight equivariant feature extraction network based on steerable convolutions under the special Euclidean group SE(2), SeS-Net, is proposed to extract overall dense features. High-response feature detection maps are generated with the dense features by an improved peakness measurement function (PMF) to derive sparse local features, which are rotation-translation invariant inherently and viewpoint-robust.
2) A local-to-global feature matching strategy is proposed, consisting of cluster-based outlier removal and adaptive high-quality match selection (AHMS). AHMS is guided by the principles of the matches' quality, spatial distribution, and regional distinctiveness.
3) An easy-to-use AVL framework featuring the steerable mechanism and the matching strategy is introduced. This framework locates the UAV within a large area autonomously and can be implemented in various scenes easily without additional training. Experiments on public image matching datasets and real UAV localization tasks demonstrate that the proposed framework exhibits good generalization, strong robustness, and high accuracy.

II. Related Works

A. Geo-Localization Based on UAV-Satellite Imagery

UAV localization based on the matching of UAV image and satellite image can be mainly divided into three categories: searching for the optimal corresponding position with template matching; employing feature descriptors constructed from points, lines, or surfaces for matching and pose estimation; and utilizing end-to-end pose estimation networks.

The methods based on template matching construct similarity metrics, such as normalized information distance (NID) [17], ROHM measure [18], and phase correlation (PC) [11], and locate the image patches with the highest similarity. These methods are robust to illumination variation but computationally expensive and sensitive to geometric deformation. Consequently, the assumption of flat terrain or orthoprojection with IMU is necessary, which is optimal for oblique image matching.

Feature-based methods involve feature detection and descriptor extraction. Feature detection aims at finding salient points or lines and constructing description vectors from neighborhoods. Handcrafted detectors such as Histogram of Oriented Gradient (HOG) [19], SIFT [7], [20], and Hessian–Laplace [21] are employed, but they failed to match images with large viewpoint changes. To handle this challenge, ASIFT [22] is adopted for aerial image matching [23], [24]. It is demonstrated in [15] and [25] that feature extraction based on deep learning is able to obtain more correct matches from images with object appearance changes. Although the method in [15] indirectly matches oblique UAV images with a series of rendered images from different angles, a high-resolution 3-D ground-truth model is required.

End-to-end deep learning methods have gained prominence in computer vision tasks in recent years, which directly estimate the transformation parameters for aligning two input images with networks [12], [26], or predict the probabilities of a sequence of discrete motion parameters with multiple reference images [16]. Such methods, however, might not consistently achieve high accuracy and exhibit limited generalization due to the lack of large public datasets.

B. Learning-Based Feature Extraction

Deep learning networks have good performance in extracting features with more semantic information from images, therefore, they are widely employed as feature descriptors, such as VGG [27], L2-Net [28], and Hard-Net [29]. They learn from positive and negative pairs of image patches extracted from the neighborhoods of the detected keypoints. To further improve the accuracy, methods such as Superpoint [30] not only describe but also detect features with neural networks instead of traditional detectors, making the two procedures cooperate better. Different from the conventional feature extraction pipeline, D2-Net [31] introduces a describe-and-detect pipeline with high efficiency and robustness, which first extracts dense features and then identifies sparse keypoints based on the feature maps.

Deep learning-based feature extractors are able to handle challenging scenes better than conventional ones, for example, they obtain more matches of close-up images and high-altitude stereo images [32]. However, addressing all differences simultaneously is difficult owing to the limited diversity of large public datasets, such as GL3D [33] and BlendedMVS [34] which mainly focus on viewpoint changes, and the aerial matching dataset [35] which only involves variations in season and illumination. UAV image matching usually suffers from drastic rotations, while CNNs are sensitive to geometric transformations such as rotation inherently. One potential solution to mitigate this issue is to implement rotation augmentation, while it requires a substantial increase in the number of parameters for the networks [36] and yields mediocre performance. In recent years, group convolutions have been introduced into image matching to gain rotation equivariance [35], [36], yielding promising results. It still leads to a significant increase in the number of parameters and induces artifacts during kernel interpolation, particularly for kernels with small sizes. Training the unconstrained group convolution network with rotation

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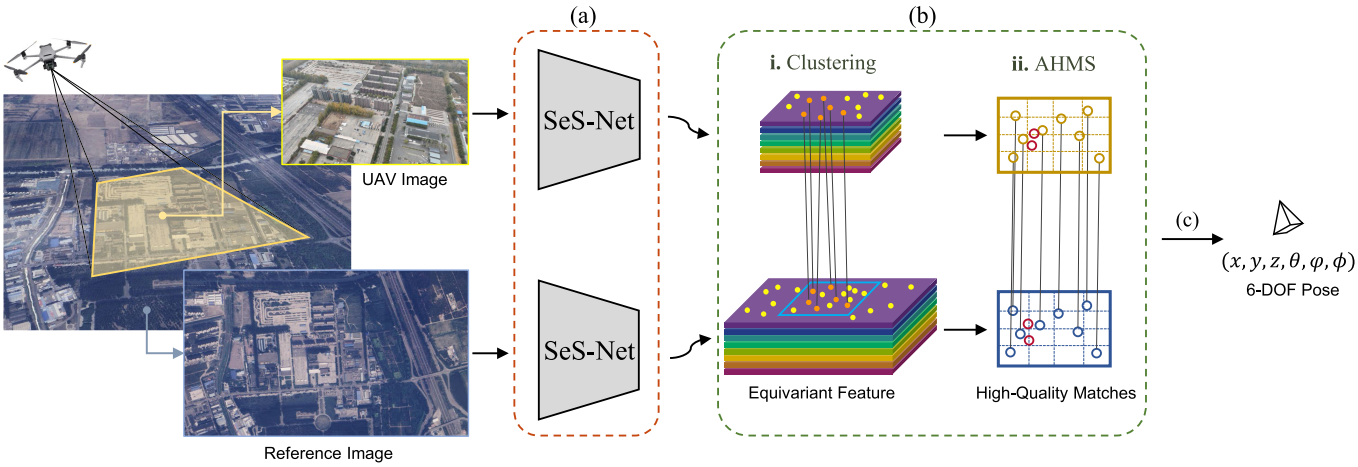


Fig. 1. Overall framework of the proposed UAV localization method. The steps in numerals (a)–(c) introduce the localization procedure. The captured UAV image and reference satellite image from one of our experiments are shown. (a) Feature extraction. (b) Local-to-Global matching. (c) PnP.

augmentation that involves interpolation on images, tends to blur the original features and decelerate the convergence.

III. Methodology

The pipeline of the proposed UAV localization method and the main components are shown in Fig. 1. First, a steerable feature extraction network, SeS-Net, with the describe-and-detect architecture extracts the rotation-equivariant features from both the UAV images and preexist offline satellite images. The UAV images of the ground surface are collected by onboard cameras at arbitrary view angles where the current position and attitude are unknown. Then, the corresponding local area in the satellite image and the valid matching pairs within are determined by clustering the initial matches based on their distribution. To further improve the accuracy of the subsequent pose estimation, the image is segmented into grid cells, and a set of high-quality matches is selected concerning several principles in the AHMS. Finally, the 6-DOF pose of the camera is estimated by the PnP algorithm with the final matches, and the UAV is geo-located. The network construction, the matching strategy, and the localization process are detailed in Sections III-A–III-C, respectively.

A. SE(2)-Steerable Feature Extraction

1) Steerable Convolution Network: The steerable convolution network is one of the implementations of equivariant networks under the Euclidean group and has been applied to image processing fields such as object detection [37]. In this article, we employ steerable convolutions to construct a lightweight feature extraction network to obtain dense feature maps from the entire input image. These dense feature maps are inherently equivariant with respect to rotation transformations as a collective entity, which indicates their invariance across every feature vector. Fig. 2 illustrates the contrast between equivariant and nonequivariant feature extraction. The nonequivariant feature vectors differ upon the rotation of the input, as do the values of detection maps generated from the feature maps. This leads to the absence and shifts of keypoints and a distinct degradation in repeatability.

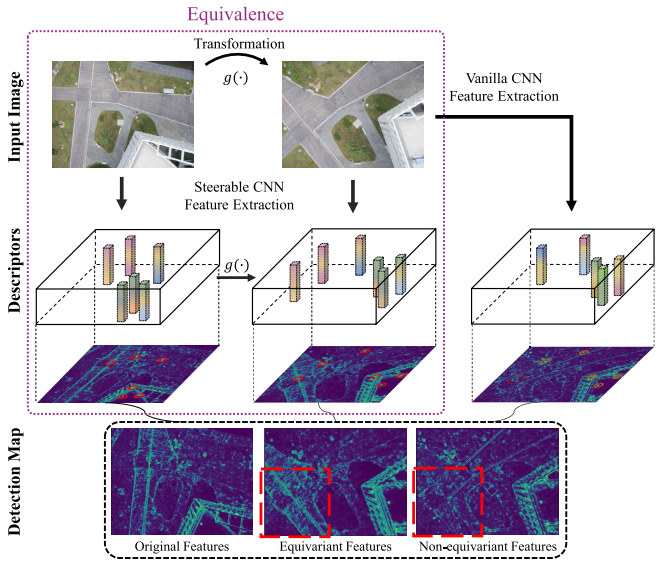


Fig. 2. Comparison between rotation-equivariant and nonrotation-equivariant networks constructed with steerable convolutions and vanilla convolutions, respectively. $g(\cdot)$ denotes the transformation and the red dashed boxes emphasize the missing feature point caused by nonequivariance.

The networks can be equivariant under certain groups with steerable convolutions. Formally, if a map f is equivariant, for any action in transformation group G

$$f(T_g \mathbf{x}) = T'_g f(\mathbf{x}), \quad g \in G \tag{1}$$

where T_g represents a group action, and T denotes a linear operation. It is indicated that the result of the mapping can be transformed in a predictable linear way under transformations of the input. Group G acts as a translation group when f represents a convolution network without supplementary constraints. In particular, T_g and T'_g are not necessarily identical, but become invariant maps when T'_g is an identity map. For $\forall(x, y) \in \mathbf{x}$, if it is shifted by T_g to the position (x', y'), and T_g is identical to T'_g, it can be inferred that the output feature vector at (x', y') is the same as $f(x, y)$ according to (1).

Steerable feature fields of steerable CNNs associate c-dimensional feature vectors $v \in \mathbb{R}^c$ with points on the

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image plane, which is similar to the feature mapping of vanilla convolutions. Feature fields are characterized by a group representation ρ which determines how they are transformed under transformations of the input. A representation of a group is a map

$$\rho : G \to GL(\mathbb{R}^n) \tag{2}$$

where $GL(\mathbb{R}^n)$ is an invertible matrix of size $n \times n$, and for the planar rotation group, $n = 2$. Regular representations are a class of representations, where the group action is permuting the channels of ρ-fields. It is well-suited for finite groups, even if it demands $|G|$ channels in each field, where $|G|$ denotes the dimension of feature fields as well as the order of the group [38]. We specify that the inputs and outputs are transformed by the action of a trivial representation that maintains the scalar values when moved. As for the intermediate feature fields passing through the network, we use regular representation.

On this basis, we construct a steerable convolution network called SeS-Net which is equivariant to rotation and translation under the special Euclidean group $SE(2)$, a subgroup of Euclidean group. Steerability implies that the transformation T'_g is independent of the input, meaning the transformed features can be steered with the pre-computed basis and coefficients. SeS-Net is implemented under the cyclic group C_N of order N, which contains N discrete rotations by angles that are multiples of $(2\pi/N)$. [38] has solved the kernel constraint that satisfies equivariant mapping of arbitrary representations, and expanded the basis into Fourier series, with the coefficients constituting the parameter space of the steerable network. With the learned Fourier coefficients ω, the feature maps extracted by a single kernel $k(x)$ can be reconstructed into multiple feature fields at certain rotation angles. The response of the steerable kernel to the input I with rotation θ is formulated as follows:

$$(I * \rho_\theta k)(x) = \sum_{q=1}^{Q} \omega(\theta)\big(I * \psi_q\big)(x) \tag{3}$$

where $\{\psi_q\}_{q=1}^{Q}$ represents the kernel basis.

The structure of SeS-Net is illustrated in Fig. 3. We utilize L2-Net [28] as the baseline, replacing the final 8×8 convolution with three 3×3 convolution layers to enhance the expressive ability and substituting convolution layers with steerable convolutions. No pooling layers are incorporated into the network to prevent the loss of rotation information. There is no limit on the size of input images, and the size of output consists of 128-dimensional dense feature descriptors which are 1/4 of the original resolution. Sparse local descriptors corresponding to specific points in the input image can be generated by interpolating the dense feature map.

Steerable convolution networks improve generalization and enhance adaptability to diverse and challenging training samples, by performing nearly ideal feature transformation that aligns with the inputs. Although training with regular representations consumes more GPU resources, the number of learnable parameters is notably lower with the hard-coded structure, and there is no extra computation time cost compared with conventional CNNs during test time.

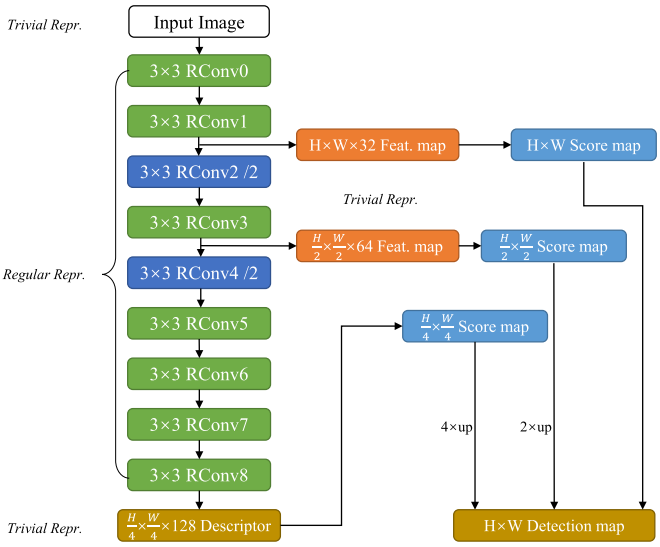


Fig. 3. Architecture of the steerable feature extraction network. RConv represents the steerable rotation-equivariant convolution, with a size of 3×3. The strides of the convolutions in green and indigo blue blocks are 1 and 2, respectively. The input image is of $H \times W$ size, and the detection map is of the same size.

2) Sparse Feature Detection: Once the dense feature map is extracted, the top K points are chosen according to their responses in the detection map. The coordinates of these keypoints are refined to subpixel accuracy through first-order Taylor expansion. Similar to ASLFeat [39], three detection score maps of different resolutions, depicted in Fig. 3, are upsampled to match the resolution of the input image and integrated by weights. The score s_{ij} of each position (i, j) in the lth score map is calculated as follows:

$$s_{ij} = \max_t \left(\alpha_{ij}^c \beta_{ij}^c\right) \tag{4a}$$

$$\alpha_{ij}^c = \mathcal{F}\left(y_{ij}^c - \frac{1}{|\mathcal{N}(i,j)|} \sum_{(i',j') \in \mathcal{N}(i,j)} y_{i'j'}^c\right) \tag{4b}$$

$$\beta_{ij}^c = \mathcal{F}\left(y_{ij}^c - \frac{1}{C} \sum_t y_{ij}^t\right) \tag{4c}$$

where $\mathcal{N}(i, j)$ represents the neighborhood around (i, j), y_{ij} denotes the output value of the convolutional layer at (i, j), and $\mathcal{F}(\cdot)$ is the PMF. The saliency with respect to the neighborhood and channel is indicated by the nonnegative scores that are mapped by $\mathcal{F}(\cdot)$, which is suggested to be a smooth activation [39]. Its characteristics have an impact on the number and positioning accuracy of keypoints, which are related to the precision of localization. We adopt Squareplus [40] as the PMF, which is defined as follows:

$$\text{Squareplus}(x, b) = \frac{1}{2}\left(x + \sqrt{x^2 + b}\right) \tag{5}$$

where b is the hyperparameter that changes the shape of Squareplus and $b \geq 0$. As shown in Fig. 4, by setting $b = 4\ln^2 2$, the shape of Squareplus approximates that of the commonly used Softplus function, and the first derivative is larger than Softplus's when $x > 0$. Leveraging these properties, SeS-Net can be trained with larger gradients in salient regions and generate higher response mapping by the PMF, which

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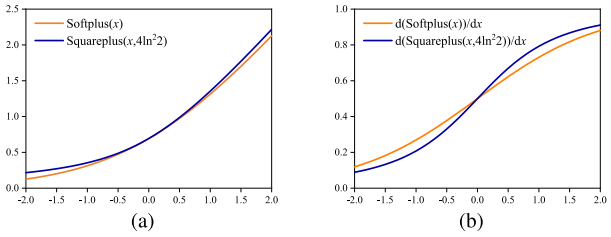


Fig. 4. Visualization of Softplus and Squareplus presented in (a), and their first derivatives presented in (b).

contributes to an increase in keypoints and an improvement in precision.

3) Loss Function: The matching loss weighted by the detection scores in D2-Net [31] is employed for training SeS-Net, which joints the learning of detection and matching, presented as follows:

$$\mathcal{L}(I_1, I_2) = \frac{1}{|\mathcal{C}|}\sum_{c\in\mathcal{C}} \frac{s_c^1 s_c^2}{\sum_{q\in\mathcal{C}} s_q^1 s_q^2} m\left(\mathbf{f}_c^1, \mathbf{f}_c^2\right) \tag{6}$$

where $\mathcal{C}$ represents the set of matches in the image pair (I_1, I_2), s_c represents the score of the matching point in the detection map, and $m(\mathbf{f}_c^1, \mathbf{f}_c^2)$ denotes the matching loss derived from feature vectors, $\mathbf{f}_c^1$ and $\mathbf{f}_c^2$, corresponding to the matching points. The matching loss takes the form of the hardest-contrastive [39]:

$$\begin{aligned} m\left(\mathbf{f}_c^1, \mathbf{f}_c^2\right) &= \left[D\left(\mathbf{f}_c^1, \mathbf{f}_c^2\right) - \alpha\right]_+ \\ &+ \frac{1}{2}\left(\left[\beta - \min_{k\neq c} D\left(\mathbf{f}_c^1, \mathbf{f}_k^2\right)\right]_+ + \left[\beta - \min_{k\neq c} D\left(\mathbf{f}_k^1, \mathbf{f}_c^2\right)\right]_+\right) \end{aligned} \tag{7}$$

where $D(\cdot,\cdot)$ denotes the Euclidean distance between two vectors, α and β are the margins of positive and negative samples, respectively, $[x]_+ = \max(x, 0)$.

B. Local-to-Global Feature Matching

Following the feature extraction with SeS-Net, we conduct the initial matching from the keypoints in the UAV image to all the keypoints in the satellite image. It may exceed the memory limit of GPUs to input the satellite image directly into the network with regard to its size. In practice, it can be divided into several nonoverlapping patches for feature extraction in batches, and the features of all batches are matched with those from the UAV image simultaneously based on L2-norm. The majority of the outliers are rejected by a relaxed consistency check of the projection model. To further remove the remaining mismatches and estimate the pose, the corresponding local area within the entire satellite image is determined and the outliers are removed by clustering the matches based on a Gaussian model. Then, a set of high-quality matches within this area is retained adaptively in consideration of the quality and quantity of the match pairs, as well as the regional distinctiveness. These accurate and well-distributed matches are the final matches for the 2-D–3-D pose estimation.

Algorithm 1 Clustering for Outlier Removal

Input: Correspondences $\mathcal{M} = \{(p_t^i, p_r^i), i = 1, 2, \ldots, K\}$;
Output: Set of inliers $\mathcal{M}'$, center of corresponding area p_c

1: **repeat**
2: $\bar{p}_r = \frac{1}{|\mathcal{M}|}\sum_i p_r^i$; #compute the initial mean of reference points
3: $\sigma = \sqrt{\frac{1}{|\mathcal{M}|}\sum_i \left\| p_r^i - \bar{p}_r \right\|^2}$; #compute the SD
#Remove the corresponding matches of the outlier points from $\mathcal{M}$
4: $\mathcal{M}' = \left\{(p_t, p_r) \middle| \left\| p_r - \bar{p}_r \right\| \le 3\sigma, (p_t, p_r) \in \mathcal{M}\right\}$;
5: $\mathcal{M}$=$\mathcal{M}'$, $p_c = \bar{p}_r$;
6: **until** $\mathcal{M}$ no longer changes.

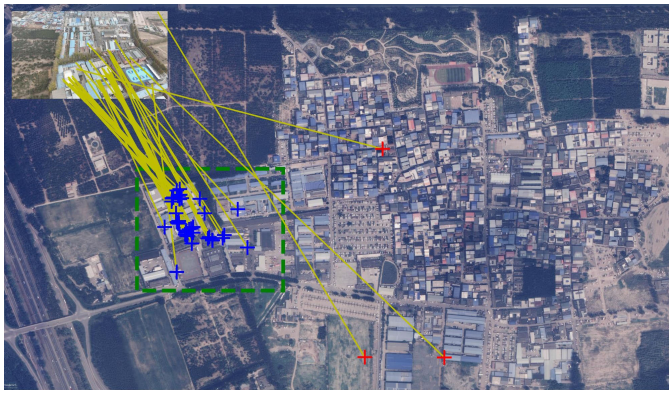


Fig. 5. Cluster-based outlier removal. The red points represent the discarded outliers, blue points represent the inliers, and green dashed box represents the corresponding area.

1) Cluster-Based Outlier Removal: Due to the more extensive ground coverage of the satellite image compared to the UAV image, the probability of finding correct matches in most regions becomes significantly low, which restricts the correct matches to a much smaller local area, shown as a dashed box in Fig. 5. The outlier removal can be cast as a clustering problem, with at least one set consisting of inliers. Assuming the positions of the matches are subject to a Gaussian distribution, the inlier set of points is defined as follows:

$$C = \{p \in D | d(p, m) \le 3\sigma\} \tag{8}$$

where D is the set of points corresponding to the candidate matches, $d(\cdot)$ denotes the Euclidean distance measurement, p represents the point in the set, m represents the current mean of the set, and σ is the standard deviation (SD) of the Gaussian model. The matching points that deviate by more than 3σ from the mean of the cluster have a low probability of belonging to the cluster, e.g., the red points in Fig. 5. In complex cases involving significant nonlinear distortion, there could be more than one cluster of inliers, while in this article, we set it as one. The procedure of the clustering is elaborated in Algorithm 1. The neighborhood with size $L \times L$ centered on p_c defines the corresponding local area, with the left-top corner serving as the georeference for the pose estimation.

2) Adaptive High-Quality Match Selection: The spatial distribution of the matched points in the image is one of

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the important factors that affect the pose estimation accuracy. UR-SIFT [41] and UC-SIFT [42] have proposed competency criteria for conventional feature detectors to extract evenly distributed features for image registration. Nonetheless, well-distributed initial points do not guarantee that the matching points remaining after the outlier removal will still be distributed evenly, especially under the circumstance of flying at low altitudes and uneven distribution of ground objects. In this case, some noise points in weak texture areas that cannot be uniquely matched are detected, resulting in the unintended removal of valid keypoints in areas with abundant features. Hence, to access a set of well-distributed matches directly, we propose a more effective strategy that involves the adaptive selection of a certain number of high-quality matches from the correct candidates.

AHMS is independent of upstream feature extractors and suitable for any detectors based on feature vectors. A larger number of candidate matches is recommended, with a suggested threshold of over 10, to prevent substantial errors. The match selection criterion is constructed mainly based on three factors.

1) Quality of matches: For high-quality correct matches, a quality measure is estimated based on the similarity of features and reprojection error. The similarity between the features of matching points is evaluated by the rank of the distance between feature vectors among all matches. The reprojection error is converted to the rank analogously to achieve a similar magnitude. The quality measure is carried out by a weighted summation of the two ranks as follows:

$$Q_i = \sum_{j=1}^{m} w^j \left(n - r_i^j\right) \tag{9}$$

where $m = 2$, n denotes the number of initial matches, r_i^j represents the ranks stated above, and w^j is the weight factor.

2) Spatial distribution of matching points: To obtain evenly distributed matching points, a grid process based on adaptive quality ranking is adopted during the local matching. An appropriate number of matches are maintained for each cell.

3) Distinctiveness of areas: To retain more stable and valid matches, the image entropy is applied to evaluate the distinctiveness of the cells. More matches in the areas with abundant objects are preserved while those in nonsalient areas are discarded.

Specifically, AHMS needs to first divide the image into multiple cells regularly [Fig. 6(a) and (b)], and then determine the number of matching points N_{Ck} within each cell considering the above factors. Finally, N_{Ck} points of the highest quality in each cell are reserved [Fig. 6(d)] and the corresponding matches participate in the pose estimation. The number of required matches is computed by the following equation:

$$N_{Ck} = N\left[\frac{w_D D_k}{\sum D_k} + \frac{w_Q \bar{Q}_k}{\sum \bar{Q}_k} + \frac{(1 - w_D - w_Q) n_k}{\sum n_k}\right] \quad k = 1, 2, \ldots, M \tag{10}$$

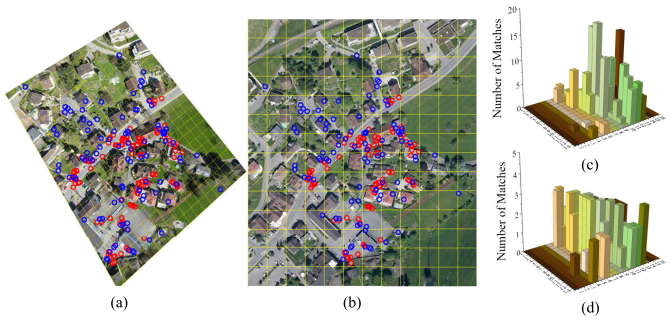


Fig. 6. Distribution of the matching points on the UAV and satellite image from Village. Blue and red points indicate the reserved and the discarded points, respectively. (a) UAV. (b) Satellite. (c) Before AHMS. (d) After AHMS.

where D_k represents the distinctiveness of the kth cell, and $\bar{Q}_k$ represents the average quality of the initial matches in the kth cell and n_k is the initial number in the kth cell. w_D and w_Q are the weight factors of distinctiveness and quality, respectively. The selection is carried out on the satellite image since the UAV images is not orthorectified. The reserved points in the UAV image are determined by the matches correspondingly.

C. Absolute UAV Localization

The PnP algorithm used for pose estimation is not limited to a particular one, and as an example, we apply the PnP optimized by Levenberg-Marquardt to solve the position and attitude of the UAV at experiments. It can be reformulated as the process of solving the initial pose with direct linear transformation (DLT), followed by the optimization of the least squares problem:

$$\xi^* = \arg\min_{\xi} \frac{1}{2} \sum_{i=1}^{n} \left\| x_i - \frac{1}{s_i} K \exp\left(\hat{\xi}\right) X_i \right\|_2^2 \tag{11}$$

where ξ represents the camera pose in Lie algebra form, K is the intrinsic matrix of the camera, x_i and X_i represent the image point and ground-truth point, respectively. Given the geo-tagged satellite image, the latitude, longitude, and altitude of the UAV can be inferred by estimating the 6-dof relative pose under the plane assumption of the ground surface. The ground truth of the UAV's localization is acquired by the onboard GPS and altimeter and converted to the UTM coordinate system.

IV. Experiments

A. Datasets

The dataset used for training is GL3D [33], which is a large general 3-D reconstruction dataset, involving 125 623 high-resolution images of 543 different scenes. The datasets for testing include a common image matching benchmark and three UAV datasets, one of which is collected by ourselves. The reference satellite images corresponding to the UAV image data are collected from Google Earth. The details of the testing datasets are as follows.

1) **HPatches** HPatches [43] is a series of 116 images of planar objects, along with the ground-truth homography.

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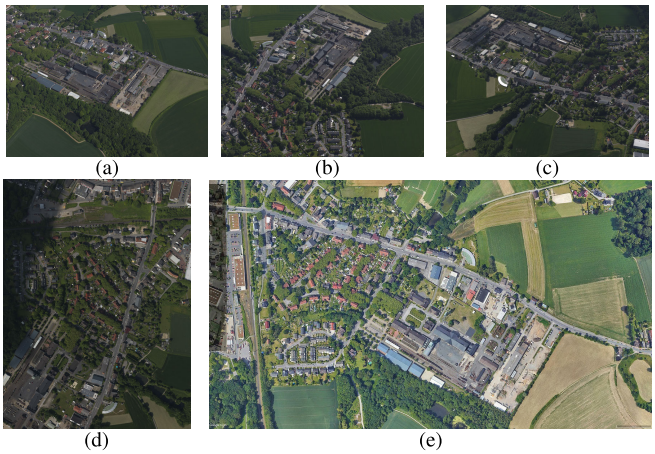


Fig. 7. UAV images of Dortmund and the corresponding reference image. The cameras collected (a)–(d) are oriented at 0°, 90°, 180°, and 270° relative to the north, respectively. (a)–(c) are oblique images with a tilt angle of 45° and view of (d) is downward. (e) Satellite image downloaded from Google Earth.

TABLE I
DETAILED INFORMATION ABOUT THE UAV FLIGHT OF CHANGPING

Flight	Length/m	Altitude/m	Tilt/°	Yaw/°	Frames	Coverage area/km^2
1	1148	169-251	17.5-60.0	-163-121	54	1.56
2	859	249-251	2.4-46.7	-76-150	41	1.56
3	1764	321-325	28.8-34.3	-174-180	35	2.7

To assess the robustness to rotations, we randomly rotated the images in HPatches within the range of 0°-180° simulating relative rotation changes, denoted as HPatches-r180.

2) **Dortmund** The ISPRS dataset [44] provides several high-resolution nadir and oblique aerial images acquired by the oblique camera system in Dortmund for image matching and dense point cloud generation. The resolution of the images is 8176 × 6132, as shown in Fig. 7(a)–(d).
3) **Village** The UAV frames from the Village dataset were captured by a nadir Canon IXUS 125 HS camera on the senseFly eBee drone [12] at about 165m height. 17 of them were chosen and the corresponding GPS-aligned satellite map was extracted from Google Earth for testing the whole localization pipeline.
4) **Changping** There are a few public real-world UAV localization datasets that account for pitch and roll variations, so we collect a dataset called Changping in the Changping district, China. The images of Changping were extracted from the videos taken by a DJI Mavic3 quadrotor during three distinct flights across various seasons and areas. These images exhibit variations in viewpoint, scale, and ground objects, and there are diverse scenes in the images including buildings, roads, grasslands, forests, and lakes. The resolution of the images is 3840 × 2160, and more details of the data acquisition are listed in Table I. The trajectories and the corresponding areas of the flights are shown in Fig. 8.

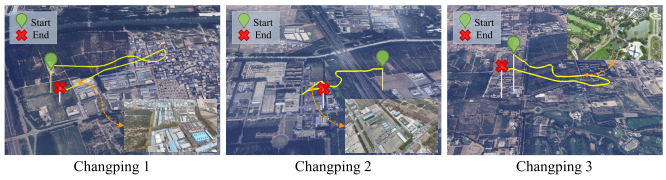


Fig. 8. Trajectories of the three flights in Changping and the examples of UAV images.

B. Implementation Detail

In this article, the experiments were conducted on Ubuntu 18.04, RTX3090 GPU, and the models were implemented based on PyTorch. The models were optimized by SGD with a momentum of 0.9 and an initial learning rate of 0.1 which decayed exponentially. The training image pairs were resized to 480 × 480 and batched 2, with the maximum match number of 512. We trained the model from scratch for 380 K iterations. The input matched grid points were determined by the ground-truth homography, and the output feature vectors were interpolated according to the sub-pixel coordinates. The experimental settings of the hyper-parameters in the matching stage are listed in Table II. It is recommended to set the size of the corresponding area to encompass the coverage of the UAV images to include more correct matches. The proposed SeS-Net has demonstrated a certain degree of robustness against scale differences, while to acquire more correct matches from images with significant scale differences, we downsampled the image and extracted features at three different scales.

The reprojection error threshold can be relaxed appropriately when RANSAC is employed, as terrain is not necessarily planar. Due to the possibility of some correct matches being omitted during global matching, it is essential to reevaluate all potential matches during the local matching. For the final step, we utilized the implementation of PnP available in OpenCV.

C. Ablation Study

1) Effectiveness and Parameter Selection of Steerable Structure: SeS-Net is capable of acquiring rotation-invariant features inherently without trained on data with rotation variation and demonstrates strong generalization ability. To verify its rotation invariance, we trained the vanilla convolution network with the same structure as SeS-Net on the training data with rotation augmented, and compared with SeS-Net. Furthermore, to determine the optimal structure of SeS-Net, we evaluated different models with different structures. In theory, the greater the number of discrete angles N of the rotation group C_N, the better the network's representation of orientations. However, an excessively large N will significantly increase the number of parameters, leading to decreased training efficiency exchanging for marginal performance improvements. [38] indicates that the improvement of handwritten letter recognition tends to saturate when $N \geq 8$, therefore in this experiment, we compared the structures with different N of 4, 8, 12. Regular representations lead to a high-dimensional feature space. For C_N, each individual feature field in the space $R^{|G|}$ consumes N channels. To ensure sufficient feature expression without massive parameters, the dimensions of feature fields, denoted as c, in the first convolutional layer were set to 8 and 12 in experiments.

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TABLE II
PARAMETER SETTINGS OF FEATURE MATCHING

Notation	Parameter	Default Value
L	Size of the corresponding area in the reference image for Village/Changping	1000/2000
N_{total}	Total number of initial keypoints from different scales	8000
s	Size of the regular grid cell	100
w^1, w^2	Weights for quality measure computation	0.5, 0.5
w^D, w^Q	Weight factors of distinctiveness and quality	0.3, 0.3

TABLE III
COMPARATIVE RESULTS OF DIFFERENT ARCHITECTURES OF THE FEATURE EXTRACTION NETWORK ON HPATCHES AND HPATCHES-R180

Dataset	Model	#Keypoints	Rep./%	M.S.	Recall/%	MMA/%	Homo. Acc./%	Param.
HPatches	Vanilla CNN	3991	51.8	33.5	**60.3**	59.5	68.3	728K
	Vanilla CNN (Rot. Aug.)	4452	53.6	27.8	48.1	55.3	67.8	728K
	$C_{4,c=12}$	4594	55.8	**33.7**	56.4	61.8	69.4	267K
	$C_{8,c=8}$	4667	**56.1**	33.4	55.5	62.2	**69.8**	**226K**
	$C_{8,c=12}$	4706	55.5	32.9	55.4	**64.6**	68.7	484K
	$C_{12,c=8}$	**4789**	55.9	32.8	54.8	62.6	69.1	323K
	$C_{12,c=12}$	4682	52.7	31.5	55.9	59.4	68.3	702K
HPatches-r180	Vanilla CNN	3860	33.6	6.7	14.3	14.0	14.4	–
	Vanilla CNN (Rot. Aug.)	4280	41.9	13.1	28.8	29.6	34.1	–
	$C_{4,c=12}$	4389	50.3	28.8	54.3	56.4	66.5	–
	$C_{8,c=8}$	4481	**51.8**	**29.6**	**54.4**	58.4	67.2	–
	$C_{8,c=12}$	**4660**	51.7	29.1	53.6	**58.8**	**67.4**	–
	$C_{12,c=8}$	4533	50.9	29.1	**54.4**	58.4	65.7	–
	$C_{12,c=12}$	4572	47.5	26.7	54.1	54.3	61.7	–

Steerable convolution models with different N and c, as well as the vanilla models, were trained and evaluated on HPatches and HPatches-r180 datasets. Certain evaluation indices are defined as follows.

1) Repeatability: The ratio of ground-truth matches to the minimum number of keypoints in the shared views.
2) Matching Score (MS) [39]: The ratio of correct matches to the minimum number of keypoints.
3) Mean matching accuracy (MMA) [31]: The ratio of correct matches and all matches.
4) Homography accuracy: The ratio of correct matched points after the reprojection with homography.

As demonstrated in Table III, the performances on HPatches of steerable models on the data without significant rotation variations are comparable to the vanilla network even if they have fewer learnable parameters, among which the number of parameters of $C_{8,c=8}$ is less than 1/3 of the vanilla's. Most steerable models exhibited higher values in most indices compared to the vanilla network except for the MS and recall, and outperforms the vanilla CNN trained with rotation augmentation except for the repeatability. In the course of experiments, we observed that the number of keypoints has some influence on the recall and MS. Specifically, fewer keypoints may lead to an increase in recall and MS. It can be inferred that steerable models with more detected keypoints obtained more valid matches than vanilla CNN actually. The results on the rotated dataset HPatches-r180 indicate that the vanilla network trained on rotation augmented data struggles to manage the rotation variations well, which yielded considerably fewer correct matches and accurate homography compared to the steerable models. It also degraded on HPatches compared to the model trained without augmentation. This suggests that SeS-Net not only learns rotation-invariant features from the original data without rotation variations but also maintained robustness to other variations in image matching, such as in illumination and viewpoint.

Among the steerable models evaluated on HPatches, $C_{4,c=12}$ and $C_{8,c=8}$ demonstrated superior performance, with the highest repeatability and homography accuracy achieved by $C_{8,c=8}$ and the best MS and recall achieved by $C_{4,c=12}$. Higher repeatability indicates more potential correct matches, and higher homography accuracy reflects accurate location of the keypoint and more valid matches, which may be beneficial to improve the precision of pose estimation. $C_{4,c=12}$ achieved higher correct ratios calculated with the number of keypoints and ground-truth matches, which indicate good distinctiveness and invariance of features, but with fewer keypoints. Model $C_{8,c=12}$ achieved more keypoints and higher MMA and homography accuracy on HPatches-r180 owing to its additional feature fields compared to $C_{8,c=8}$. However, $C_{8,c=8}$ attained higher repeatability, MS and recall, with less than 1/2 the number of parameters. Considering the performance across these two datasets, the subsequent experiments were conducted based on model $C_{8,c=8}$.

2) Comparison of PMF: The computation of keypoint responses requires a smooth PMF, which can also prompt potential matching points to achieve higher responses while suppressing the responses in flat areas. Squareplus has the advantages of easy computation, numerical stability, and feature learning compared to Softplus. We compared these two

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TABLE IV
COMPARISON OF DIFFERENT PMFS ON HPATCHES

PMF	Rot. HP	#Keypoints	Rep./%	M.S.	Recall/%	MMA/%	Homo. Acc./%
Softplus	✗	**4717**	56.0	32.9	55.0	62.2	69.6
	✓	**4506**	50.8	28.6	53.5	57.6	65.9
Squareplus	✗	4667	**56.1**	**33.3**	**55.5**	**62.2**	**69.8**
	✓	4481	**51.8**	**29.6**	**54.4**	**58.4**	**67.2**

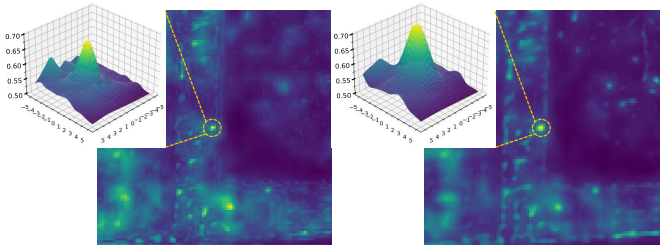


Fig. 9. Detection heatmaps at the keypoint extracted by SeS-Net with Squareplus and Softplus.

functions based on the $C_{8,c=8}$ model of SeS-Net. Table IV shows that Squareplus was able to match better despite a slightly smaller number of keypoints than Softplus, particularly evident on HPatches-r180. It demonstrates that the keypoints detected with Squareplus for matching are more salient and the positions of keypoints are more precise. The localization errors of Squareplus were lower than Softplus generally, as indicated in Table V, except for the height error on Changping 2. Nonetheless, Squareplus showed a certain level of superiority over Softplus in terms of localization due to the emphasis on horizontal localization accuracy. We visualized the response map of detection around the extracted keypoint, as shown in Fig. 9. It can be observed that the spike at the keypoint with Squareplus is higher and sharper, while the values nearby are lower compared to Softplus's. With higher responses at salient points, it is easier to obtain more valid keypoints, and the noise points are less likely to be detected.

3) Effectiveness of AHMS: The purpose of AHMS is to select representative matching points with high quality and well-distributed from the densely distributed areas for pose estimation, and ignore the matches with less importance and reliability. An example of matches with adjusted distributions is displayed in Fig. 6. Matches in the featureless area were preserved, while the matches in other areas were reduced adaptively, resulting in an approximately uniform distribution of matches across different cells after AHMS. Furthermore, we compared and analyzed the number of matches and the localization error of the proposed method with and without this strategy. The results are presented in Table V. The accuracy of the localization was improved after employing the strategy, even if the number of matches was reduced. This can be attributed to the fact that many matching points are concentrated on the same objects, resulting in an overemphasis on reducing the errors of the dense matches rather than considering the overall errors during pose estimation. If the matching points are concentrated on the areas with nonlinear deformation, it may even introduce more errors.

D. Comparative Experiments

1) Image Matching Evaluation: SeS-Net can be applied to the matching between UAV images and satellite images directly after training on GL3D without fine-tuning. To assess the robustness of the features extracted by SeS-Net, a matching test was conducted on the Dortmund dataset with various feature extractors, such as SIFT [45], ASIFT [22], GIFT [36], and ASLFeat, for comparison with SeS-Net. GIFT [36] is a group convolutional network claimed to have rotation invariant. The best model of ASLFeat provided by the authors and equipped with unconstrained deformable convolutions is sensitive to rotation. Therefore, we fine-tuned it to enhance its rotation robustness, and the refined model is referred to as ASLFeat-Rot.

The number of correct matches and the correct match ratio (CMR) are presented in Fig. 10. The k-nearest neighbors algorithm and RANSAC were utilized in the matching procedure. It is shown that SeS-Net was able to acquire more correct matches under multiple circumstances such as large tilt and yaw angles of UAV and drastic illumination variation. Besides, the precision was significantly higher for all pairs of images except the first one, where more than 40 correct matches were obtained. Although SIFT and ASIFT could resist rotation and viewpoint changes to some extent, they failed easily when the angles were too large or faced multiple variations. Specifically, SIFT showed certain repeatability when dealing with rotational changes alone such as pair (d), whereas was likely to fail with respect to oblique images. ASIFT was capable of handling certain cases such as pair (a) and (c) which involve tilt, while struggled when the images are under asymmetric or severe rotation changes, such as (b) and (d). ASLFeat-Rot strengthened the robustness of rotation and could adapt to multiple changes in viewpoint, while the number and ratio of correct matches were low. GIFT only exhibited a low false match rate when achieving certain correct matches successfully. However, the correct matches were few, which makes it more vulnerable than other methods when encountering severe tilt and rotation variations due to the lack of generalization.

Among the feature extraction methods based on deep learning, the learnable parameters of SeS-Net are notably fewer than the others, as shown in Table VI, totaling just 226 K. ASLFeat increases the number of parameters and slows down the computation distinctly by incorporating deformable convolutions. The conventional detect-and-describe pipeline of GIFT is inefficient and contains numerous redundant parameters. The most superior performance achieved by SeS-Net with fewer parameters is not only attributed to a well-designed structure and effective training methods, but it is also a result of introducing steerable convolutions. Once the coefficients are learned, the kernel is required only once and can be reused during the evaluation, which is beneficial to low-cost and real-time localization.

2) UAV Localization: The whole pipeline of the proposed method for UAV localization was evaluated on the Village and Changping datasets. We adopted the same PnP algorithm after feature extraction and matching for the point-matching-based methods. The assessments were quantified by the absolute errors in horizontal and vertical directions. Due to the large

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TABLE V
EFFECTIVENESS OF IMPROVED PMF AND AHMS

Improv. PMF	AHMS	Village			Changping1			Changping2			Changping3		
		#Matched	$\overline{\lvert\Delta XY\rvert}$/m	$\overline{\lvert\Delta Z\rvert}$/m	#Matched	$\overline{\lvert\Delta XY\rvert}$/m	$\overline{\lvert\Delta Z\rvert}$/m	#Matched	$\overline{\lvert\Delta XY\rvert}$/m	$\overline{\lvert\Delta Z\rvert}$/m	#Matched	$\overline{\lvert\Delta XY\rvert}$/m	$\overline{\lvert\Delta Z\rvert}$/m
		239.1	6.69	6.65	80.2	8.60	10.40	105.6	6.35	**3.91**	71.4	11.18	15.54
✓		231.4	6.58	6.48	78.7	8.42	10.33	116.7	5.89	4.30	83.6	8.16	15.31
✓	✓	106.9	**5.52**	**6.20**	30.9	**7.90**	**9.87**	43.4	**5.56**	4.19	33.3	**7.51**	**15.23**

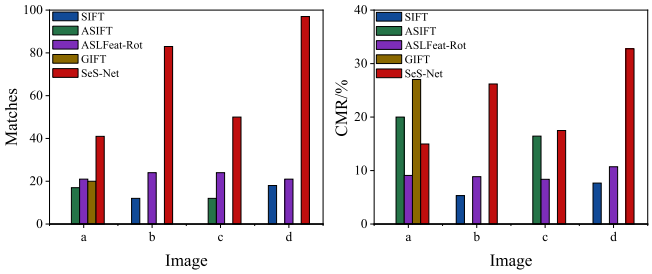


Fig. 10. Number and ratio of correct matches on Dortmund. The bars in red represent our method.

TABLE VI
COMPARISON OF THE PARAMETERS OF FEATURE EXTRACTION NETWORKS

Model	ASLFeat	GIFT	SeS-Net
Params	822K	1301K+424K	**226K**

TABLE VII
AVERAGE LOCALIZATION ACCURACY ON VILLAGE

Method	#Matches	$\overline{\lvert\Delta XY\rvert}$/m	$\overline{\lvert\Delta Z\rvert}$/m
SIFT	13.3	16.44	7.75
ASIFT	19.8	26.1	17.8
ASLFeat-Rot	90.9	32.62	11.61
GIFT	13.41	\	\
ICLK-PE	\	7.06	8.01
Ours	**106.9**	**5.52**	**6.20**

magnitude of coordinates under UTM projection, we computed the localization errors with respect to the corresponding local areas to make the pose solving process more stable.

Apart from the methods mentioned above, we compared the localization results of ICLK-PE [12] on Village with our method. The comparison results on Village are presented in Table VII. The images in Village were taken from a nadir camera, which provided a view angle similar to that of the satellite image, therefore, the conventional methods such as SIFT and ASIFT were able to fulfill the localization in most cases but with minimal matches, which resulted in high errors and unreliable positions, especially in horizontal direction. SIFT achieved higher average accuracy than ASIFT because of fewer failures and better adaptability for images without viewpoint changes. Similar to the performance on Dortmund, GIFT could hardly locate the UAV successfully due to the lack of correct matches in most frames. ASLFeat-Rot could get more matches than the methods above, but like ASIFT, it failed to locate in some cases when matched features are insufficient. The errors became intolerable when the count of valid matches falls below about 10. As demonstrated in [12], ICLK-PE has achieved the localization precision of 7.06 and 8.01 m in horizontal and vertical directions, respectively. Our proposed method is capable of obtaining a large number of matches among limited ground features compared to other methods and achieved the highest number of matches along with the lowest horizontal localization error of 5.52 m and height error of 6.20 m.

Fig. 11 shows the localization results of the proposed method and the others for the image frame sequences of the Changping dataset, while Fig. 12 displays the corresponding attitudes estimated by the proposed method. Notably, SIFT and GIFT failed to locate in most cases on Changping which is mainly composed of oblique images, and could only succeed in the mid-period of the second flight. ASIFT achieved localization without significant errors only in the second and third flights. It can be indicated from Fig. 12 that under moderate three-axis viewpoint changes, such as attitude angles less than 60° in frames 18–27 of the second flight, all methods, particularly SIFT, consistently yielded reliable location information, as shown in the corresponding curves in Fig. 11. The methods mentioned above tend to fail when confronted with large tilt (involving pitch and roll) or yaw angles. This tendency is particularly evident in the frames surrounding the 30th frame during the first flight and the final 28th–35th frames of the third flight. ASLFeat-Rot was capable of getting reliable positioning information in most cases, but the overall accuracy was inferior to the proposed method. It also encountered instances where the location occasionally failed due to the lack of matches.

To further investigate the overall performance on Changping for the competitive methods that are able to obtain valid results, the average position error and its SD were calculated and analyzed for ASIFT, ASLFeat-Rot, and our method. The SD serves as an indicator of localization stability. It is evident in Fig. 13 that ASIFT was only able to achieve stable localization on the second flight with the largest error among them. ASLFeat-Rot maintained consistent localization accuracy during the second and third flights, with the SDs both below 20 m. It exhibited smaller errors compared to ASIFT, while the mean errors exceeded 10 m except for the second flight. The proposed method achieved the highest accuracy across the three flights, with minimum average horizontal and height errors of 5.34 and 4.50 m, respectively. The position errors remained under 10 m except for the height error in the third flight. In addition, we attained the smallest error SDs,

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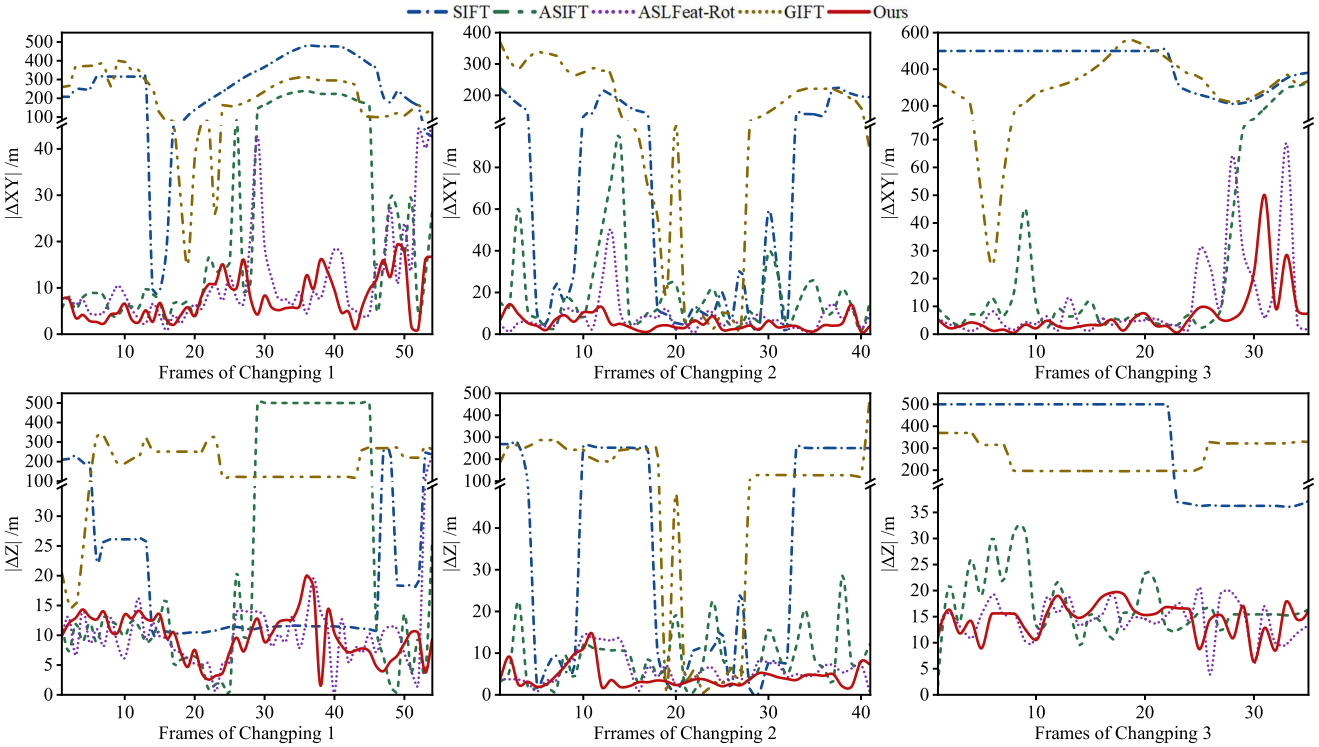


Fig. 11. Average horizontal localization and height errors estimated by different methods on Changping.

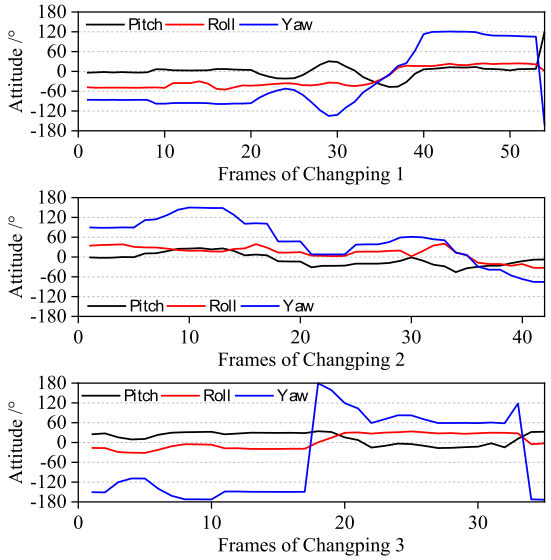


Fig. 12. Attitudes of the data in Changping.

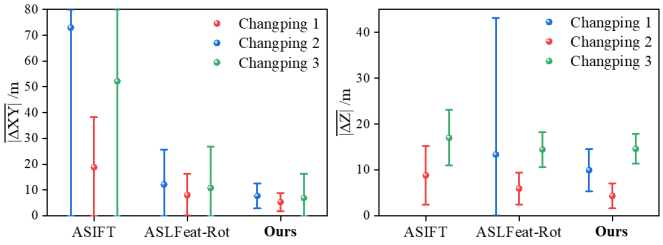


Fig. 13. Statistics of horizontal position and height errors. The data points denote the average errors, $\bar{e}$, and error bars show the interval of $(\bar{e} \pm \sigma)$.

all of which were below 10 m and the minimum was 2.6 m, indicating the highest stability and infrequent localization failures. In general, the proposed method outperformed other methods not only on the nadir-view data such as Village, but also on the three flights of the Changping dataset with severe viewpoint changes.

To reveal the beneficial impact of the match selection strategy in the proposed localization pipeline on improving the localization accuracy, AHMS was compared with the recent outlier removal methods, LPM [46] and GLOF [47], which have shown good performance on stereo matching. The features for matching were detected and extracted by SeS-Net. In addition to the metrics used for localization evaluation, the local distribution density [48] is adopted to assess the distribution of the matches, which is rewritten as:

$$\delta_{dens} = \frac{\sum_{i=1}^{N} \left| A_i - \overline{A} \right|^2}{\sum_{i=1}^{N} A_i} \tag{12}$$

where A_i denotes the ith Voronoi cell, $\overline{A}$ represents the mean area of all Voronoi cells cut by the image boundary. It signifies the disparity in distances between points. A low density indicates a more evenly distributed pattern.

The matching and localization results are shown in Table VIII. AHMS achieved the best performance with significantly lower localization error and distribution density than other methods, except on the Village dataset. The localization errors of GLOF were close to ours on Village due to the large number of matched points it obtained, while the distribution was inferior. LPM acquired more points than ours but yielded

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TABLE VIII
COMPARATIVE RESULTS OF DIFFERENT MATCH SELECTION STRATEGIES

		LPM	GLOF	AHMS
Village	#Matched	46.8	**214.9**	106.5
	δ_{dens}	0.550	0.592	**0.532**
	$\overline{\lvert\Delta XY\rvert}$/m	17.60	6.71	**5.52**
	$\overline{\lvert\Delta Z\rvert}$/m	6.06	**5.35**	6.20
Changping 1	#Matched	60.7	**78.3**	30.9
	δ_{dens}	0.480	0.542	**0.385**
	$\overline{\lvert\Delta XY\rvert}$/m	16.05	14.45	**7.90**
	$\overline{\lvert\Delta Z\rvert}$/m	23.03	11.32	**9.87**
Changping 2	#Matched	102.4	**109.1**	43.4
	δ_{dens}	0.503	0.533	**0.482**
	$\overline{\lvert\Delta XY\rvert}$/m	21.53	11.63	**5.56**
	$\overline{\lvert\Delta Z\rvert}$/m	9.24	5.43	**4.19**
Changping 3	#Matched	**110.9**	49.3	33.3
	δ_{dens}	0.588	0.441	**0.379**
	$\overline{\lvert\Delta XY\rvert}$/m	57.43	34.34	**7.51**
	$\overline{\lvert\Delta Z\rvert}$/m	20.34	23.39	**15.23**

the poorest localization performance disregarding the local density. Although GLOF was able to match more points, not all of these matches contributed positively to enhancing accuracy. It can be inferred that LPM and GLOF struggle to estimate reliable positions when there are substantial variations in viewpoint, as evidenced by the high average error on Changping 3 dataset. The high-quality matches preserved after filtering by AHMS were well distributed and contributed to successful localization in all scenarios.

The detected and matched feature points are salient keypoints discarding the background information. They do not lie on the same horizontal plane when situated on buildings or hills in practice, which causes certain pose estimation errors under the plane consumption. These points with significant elevation need to be identified and eliminated to achieve higher precision, such as sub-meter accuracy. Thus, we will study image matching and localization with semantic information in the future.

V. CONCLUSION

In this article, we summarized the limitations of previous UAV localization methods based on UAV-satellite image matching and proposed a visual localization method for GPS-denied UAV to address the challenges of oblique UAV image matching. Most of the existing methods based on feature extraction address single or limited variations, which have difficulties accommodating the UAV-satellite image matching characterized by severe changes in rotation, viewpoint, ground surface, etc. To this end, the feature extraction network, SeS-Net, designed in this article adopts steerable convolutions under the rotation group and the describe-and-detect mechanism. With a lightweight structure, SeS-Net surprisingly excels at extracting features robust to multiple variations, particularly the rotation and viewpoint changes. An improved peakness measurement which results in high responses and quick convergence was introduced to generate sparse feature points. To further improve the localization accuracy, a local-to-global feature matching strategy was proposed, which removes the outliers based on clustering and reserves the high-quality matches within the corresponding local area based on the distribution, quality, and distinctiveness. The extensive experiments demonstrate that the proposed SeS-Net and the localization framework are superior to both conventional and deep learning methods with strong generalization. The proposed method achieves average horizontal and altitude accuracies of 5.52 and 6.20 m on the Village dataset, and the minimum errors of 5.56 and 4.19 m on the Changping dataset with oblique images. The localization of our method is the most stable with a minimum SD of 2.6 m. The plane consumption introduces certain errors due to the elevation. Therefore, it can be a direction to improve accuracy by involving semantic information for feature extraction.

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